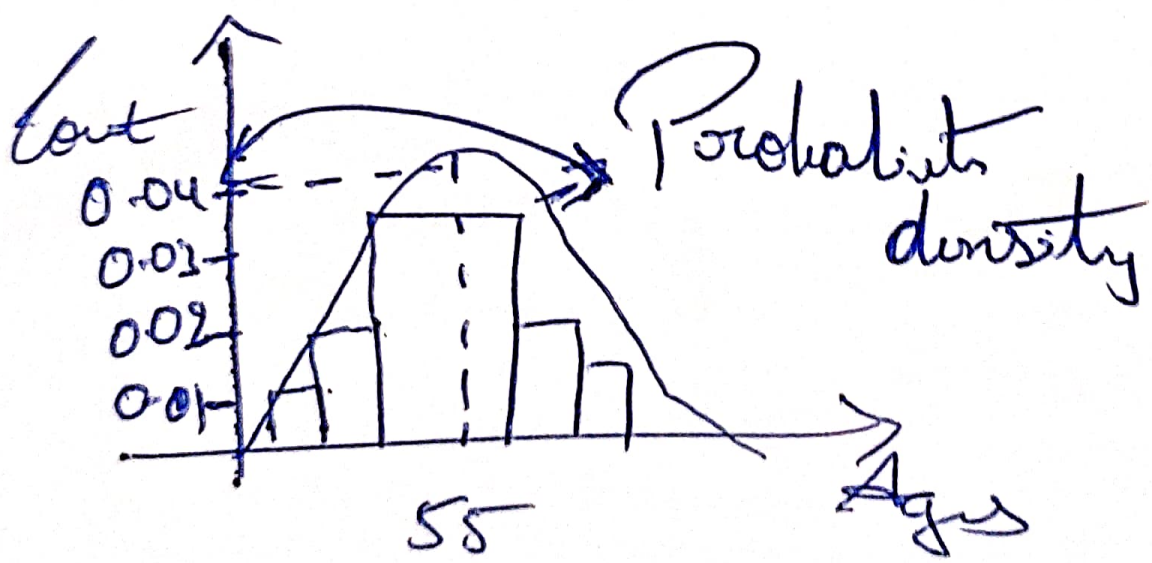
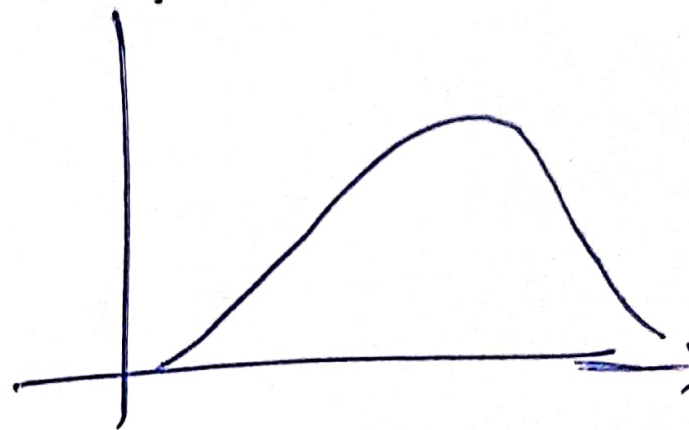
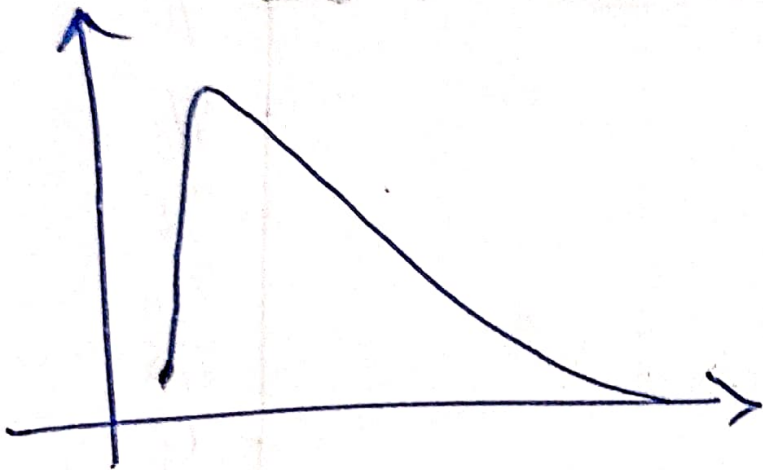
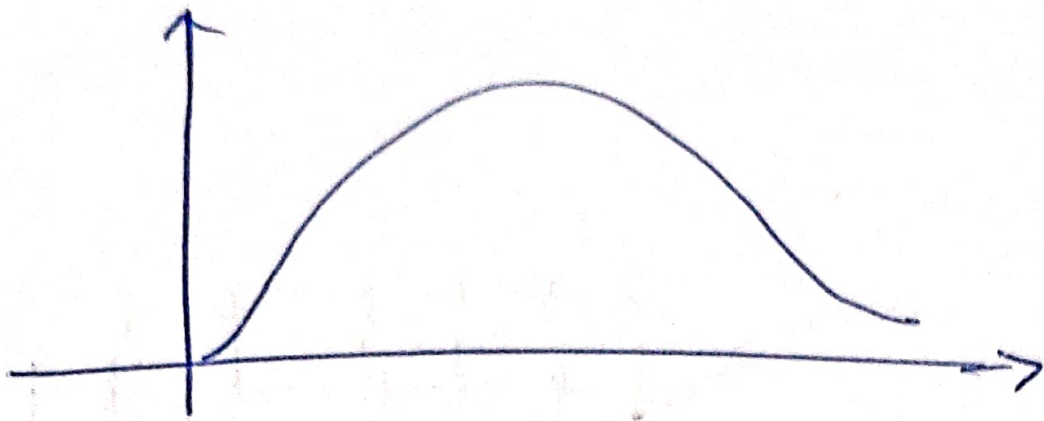


Probability Distribution Functions

Probability distribution functions describe how the probabilities are distributed over the value of a Random Variable.

Age = 8 - - - - - } $\Rightarrow$ Continuous random Variable





3 Main of probability distribution functions

① Probability Mass function (PMF)
used for discrete random variable

② Probability density function (PDF)

used for continuous random variable

③ Cumulative Distribution function (CDF)

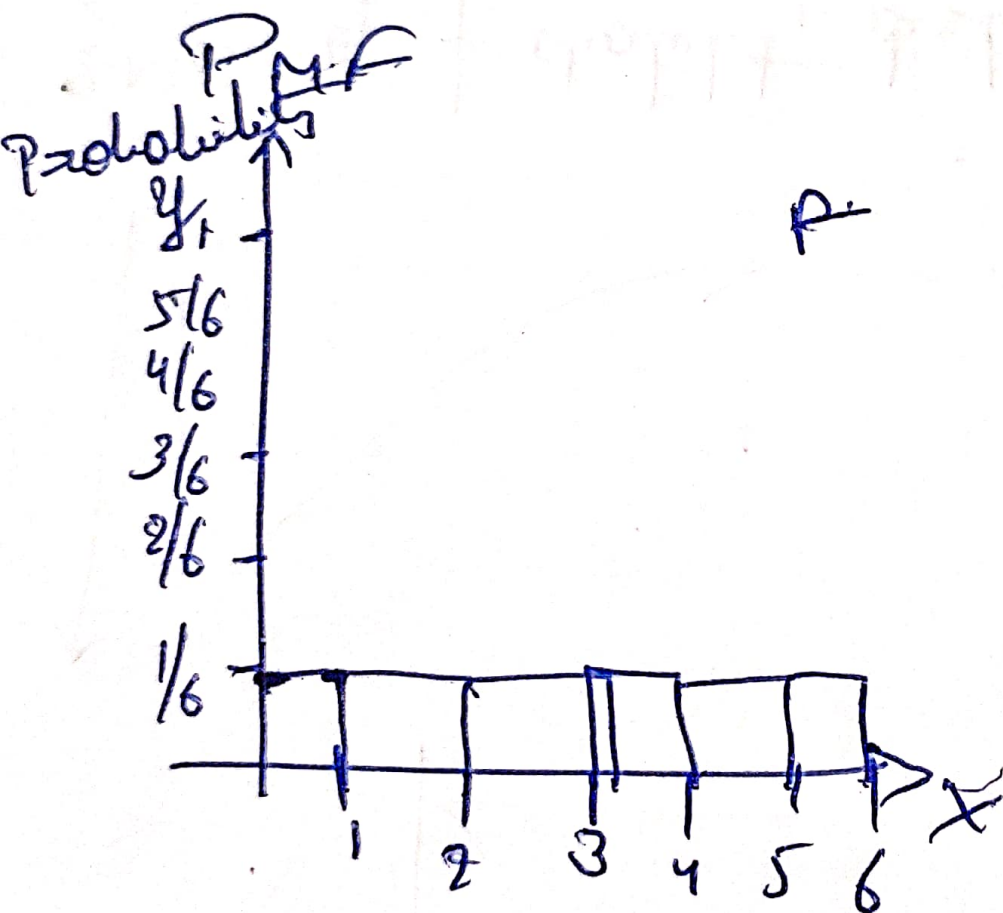
① Probability Mass Function

[Discrete Random Variable]

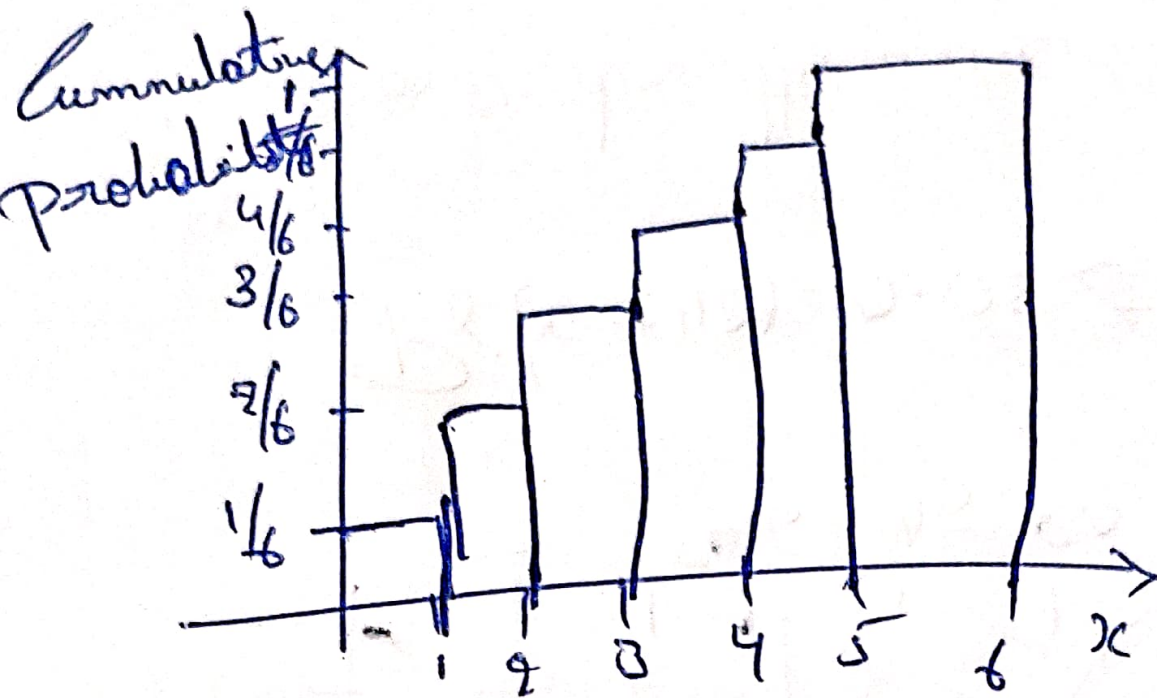
Eg:- Rolling a dice $\{1, 2, 3, 4, 5, 6\}$

↓
Fair Dice

$$P_X(1) = P_X(2) = P_X(3) = P_X(4) = P_X(5) = P_X(6) = \frac{1}{6}$$



Cumulative Density Function (cdf)



Ex

$$P(x \leq 2) = P(x=1) + P(x=2)$$

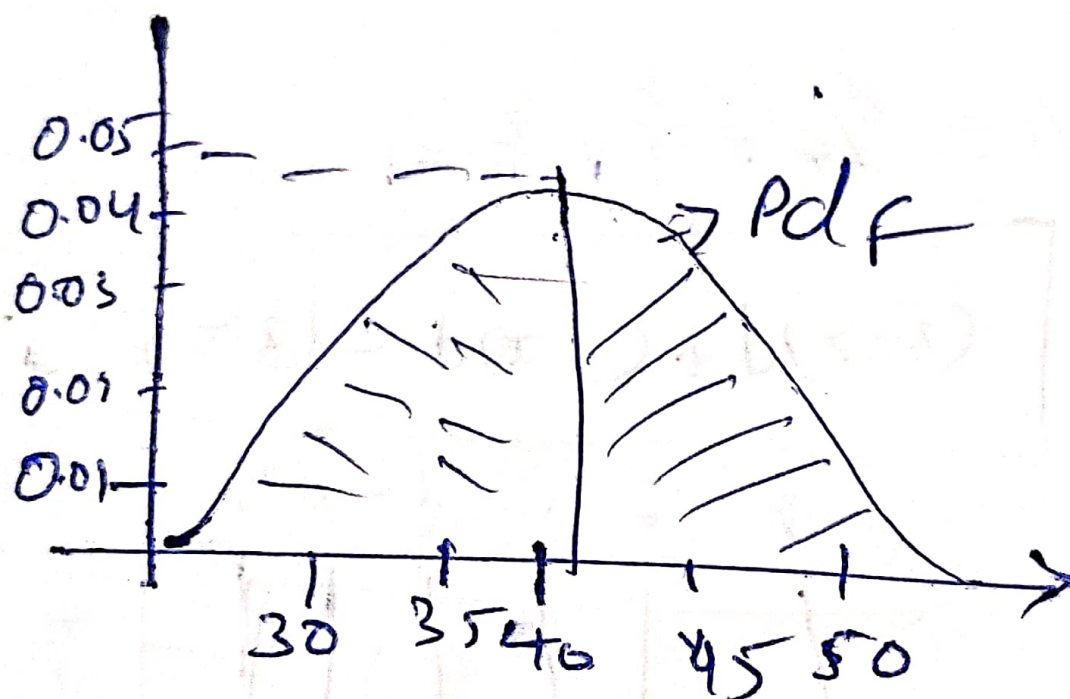
$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

$$P(x \leq 6) = P(x=1) + P(x=2)$$

② Probability Density Function (pdf)

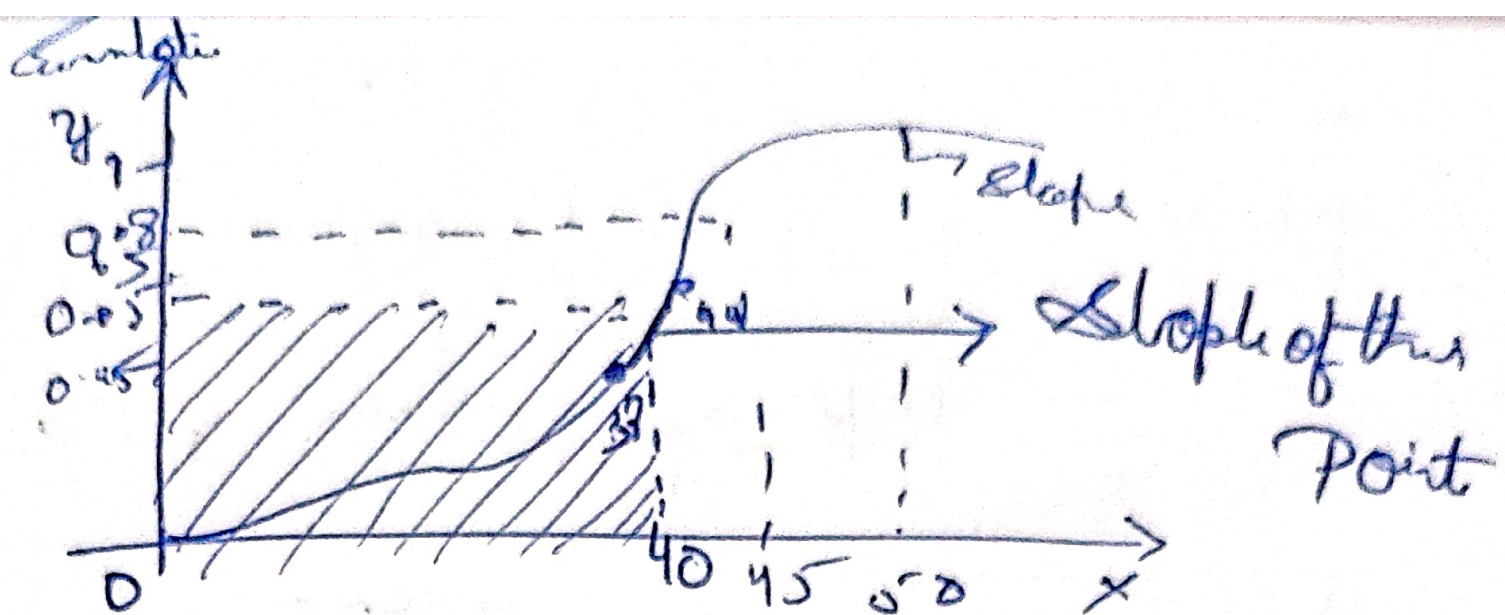
① Distribution of Continuous Random Variable

$$x = \text{Ages} = \{ \dots \}$$



$$P(x = 40) = 0.05 \Leftarrow$$

- ① Area under the curve
- ② probability density



$$P(X \leq 40) \rightarrow 0.5 \Rightarrow 50\%$$

Slope $\uparrow$ Density $\uparrow$

$$\text{Slope} = 44.$$

$$= \left\{ \frac{0.55 - 0.45}{44 - 38} \right\} \Rightarrow 70\%$$

(x_1, y_1)
 (x_2, y_2)

$$\text{Slope} = \left[\frac{y_2 - y_1}{x_2 - x_1} \right]$$

$\Leftarrow$
Gradient $\Rightarrow$ probability percent

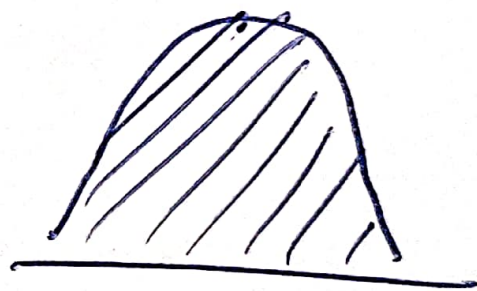
Probability Density $\Rightarrow$

Gradient of Cumulative density function.

Properties of PDF

- ① Non-Negativity $f(x) \geq 0$ for all x
- ② The total area under the PDF curve is Equal to 1

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



with respect to different distribution $f(x)$ function is going to change.